MODEL QUESTION PAPER

Program Name:Automobile Engineering **Semester: Four**

Course code: **Course name: HEAT POWER ENGINEERING**

Time : 3 Hours *Max.Marks : 75*

I. **Answer all questions in one word or one sentence** **(9 x 1 = 9 Marks)**

1	Thermal flask is an example of ---------------- system	M 1.01	R
2	The Unit of specific heat at constant volume is-----------------	M 1.01	R
3	What is cut off ratio?	M2.05	R
4	Define stroke length?	M 2.01	R
5	The ideal cycle for Diesel engine is------------------	M 2.05	R
6	Relative efficiency is also known as-------------------ratio.	M 3.01	R
7	The brake power of an engine is always -------------the indicated power.	M 3.04	U
8	The multistage air compression of air as compared to single stage air compression a) Improves volumetric efficiency b) Reduces workdone per kg of air c) Gives more uniform torque. d) All of the above.	M 4.05	U
9	Fourier's law is the rate equation for -------------------	M 4.02	R

II. **Answer any Eight questions from the following** **(8 x 3= 24 Marks)**

1	Express in words the conservation of energy principle for a closed system.	M 1.02	U
2	What do you understand by a thermodynamic system	M 1.01	R
3	Explain how thermodynamic cycles are classified.	M 2.01	U
4	Outline any four assumptions in thermodynamic cycle analysis.	M 2.01	U
5	Compare PV diagrams of Otto and Carnot cycle.	M 2.03 M2.04	U

6	What is the effect of volumetric efficiency on engine power and specific fuel consumption?	M 3.01	U
7	List out all items in a typical heat balance sheet.	M 3.04	R
8	How does forced convection differ from natural convection?	M4.03	R
9	What are the advantages of multi stage air compression?	M 4.05	R
10	Summarize Stefan-Boltzmann law of total radiation.	M 4.02	U

III. **Answer all questions from the following (6x 7 = 42 Marks)**

1	A piston cylinder containing air expands at a constant pressure of 150 kPa from a temperature of 285 K to a temperature of 550 K. The mass of air is 0.05 kg. Solve (a) heat transfer (b)work transfer and (c) change in internal energy during the process. Take Cp= 1 kJ/kg K and R=0.287 kJ/kg K.	M 1.05	A
	OR		
2	A quantity of air has a volume of 0.0565 $m^3$and a pressure of 7.03 bar. It is expanded in a cylinder to a pressure of 1.05 bar. Apply appropriate equations and compute the work done if the expansion is (1). Hyperbolic (2). Adiabatic .($\gamma = 1.4$)	M 1.05	A
3	A quantity of gas has a volume of 0.14 m^3, pressure 1.5 bar and a temperature 100 °C. If the gas is compressed at constant pressure, until its volume becomes 0.112 m^3, Utilize the equations and determine temperature at the end of compression and work done in compressing the gas. Take Cp = 1.005 kJ/kg K &Cv= 0.712k J/kg K and R=285 J/kg K.	M 1.05	A
	OR		
4	A quantity of gas is compressed according to $PV^{1.25}$ = constant. The initial temperature and pressure of the gas is 15 ^oC and 1bar respectively. Make use of the equations and determine the work done in compressing 1kg of air at 3 bar and heat rejected through the walls of the cylinder. γ for air as 1.4, R= 311.4J/kgK.	M 1.05	A
5	In an ideal Otto cycle the air at the beginning of of isentropic compression is at 1 bar and 15^0C. The ratio of compression is 8. If the heat added during the constant volume is 1000kJ/kg, Solve maximum temperature in the cycle and the air standard efficiency. Take γ for air as 1.4. and Cv = 0.718 kJ/kgK	M 2.04	A

	OR		
6	An amount of a perfect gas has initial conditions of volume 1 m^3. pressure 1 bar and temperature 18^0C. It undergoes ideal diesel cycle operations, the pressure after isentropic compression being 50 bar and the volume after constant pressure heat addition being 0.1 m^3. Apply the equations of Diesel cycle and find out the temperatures after adiabatic compression and end of constant pressure heat addition .γ for air as 1.4.	M 2.05	A
7	Explain the method of calculation of brake power using rope brake drum dynamometer.	M 3.01	U
	OR		
8	Outline the steps for determining specific fuel consumption	M 3.01	U
9	A gas engine working on 4 stroke constant volume cycle gave the following results during a test of an hour's duration. Heat supplied by the fuel = 10280 kJ/min, indicated power=20.8 kW, brake power=18400 W, mass of cooling water circulated = 660 kg/h. cooling water temperature rise =34.2^0C. Heat loss to exhaust gas = 8 %. Model a heat balance sheet for the engine.	M 3.05	A
	OR		
10	A gasoline engine working on four stroke develops a brake power of 20.9 kW. A Morse test was conducted on this engine and brake power (kW) obtained when each cylinder was made inoperative by short circuiting the spark plug are 14.9, 14.3, 14.8, and 14.5 respectively. The test was conducted at constant speed. Solve indicated power and mechanical efficiency.	M 3.03	A
11	Explainthe applications of heat transfer in Automobile Engineering Field.	M 4.01	U
	OR		
12	Compare Fourier's law and Newton's law of cooling.	M4.02	U

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Mark Distribution

Module	Hours/Module (hi)	Marks/Module $(h^i/\sum H^i) * 123$ (±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No. of questions	Marks	No. of questions	Marks	No. of questions	Marks	No. of questions	Marks
1	15	36	2	2	2	6	4	28	8	36
2	14	26	3	3	3	9	2	14	8	26
3	15	36	2	2	2	6	4	28	8	36
4	14	25	2	2	3	9	2	14	7	25
Total	58	123	9	9	10	30	12	84	31	123

Cognitive Level Distribution

Cognitive Level	**Marks**	**% of Marks**
Remembering	19	16
Understanding	48	39
Applying	56	45
Total	123	100